DATABASE MANAGEMENT LAB

LAB Report

Submitted in partial fulfilment of the requirement of the degree of

BACHELORS OF TECHNOLOGY

to

K.R Mangalam University

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SESSION – 2025-2026

Experiment : 1

Question 1. Consider following databases and draw ER diagram and convert entities

and relationships to relation table for a given scenario:

COLLEGE DATABASE

STUDENT (USN, SName, Address, Phone, Gender)
SEMSEC (SSID, Sem, Sec)
CLASS (USN, SSID)
SUBJECT (Subcode, Title, Sem, Credits)
IAMARKS (USN, Subcode, SSID, Test1, Test2, Test3, FinalIA)

Solution:

1. The entities **STUDENT, SEMSEC, SUBJECT, and IAMARKS** are identified and represented using rectangles.
2. The attributes are represented using ellipses:

o **STUDENT:** USN (Primary Key), SName, Address, Phone, Gender
♣ USN is underlined as it is the Primary Key.
♣ Phone is shown as a single-valued attribute.

o **SEMSEC:** SSID (Primary Key), Sem, Sec
♣ SSID is underlined as the Primary Key.

o **SUBJECT:** Subcode (Primary Key), Title, Sem, Credits
♣ Subcode is underlined as the Primary Key.

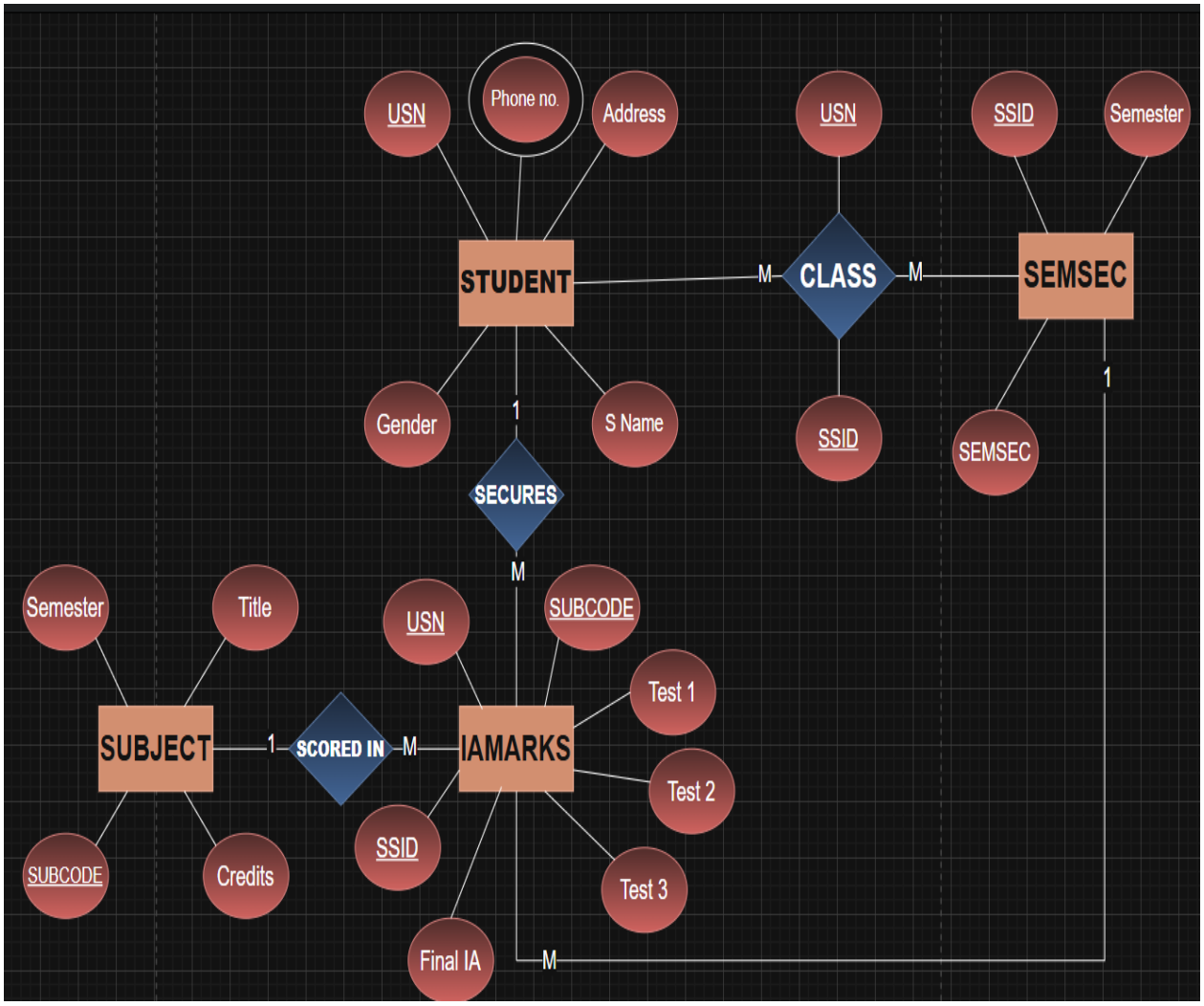
o **IAMARKS:** USN, Subcode, SSID, Test1, Test2, Test3, FinalIA
♣ USN, Subcode, and SSID together form a composite key.
♣ Test1, Test2, Test3, and FinalIA are normal attributes.

3. Primary keys are represented by underlining the attribute names.
4. **CLASS** is drawn as a relationship (diamond shape) between **STUDENT** and **SEMSEC**.
5. The relationship between **STUDENT and SEMSEC** through CLASS is **M : 1**, meaning many students belong to one semester and section.
6. The **M : N relationship between STUDENT and SUBJECT** is resolved using the associative entity **IAMARKS**.
7. The relationship **STUDENT – IAMARKS** is **1 : M**, indicating one student can have multiple internal assessment records.
8. The relationship **SUBJECT – IAMARKS** is **1 : M**, indicating one subject can have multiple IA records for different students.

9. The relationship **SEMSEC – IAMARKS** is **1 : M**, showing that marks are maintained semester-section wise.

10. All entities, attributes, relationships, primary keys, foreign keys, and cardinalities are clearly represented in the ER diagram.

ER DIAGRAM:



Experiment : 2

Question 2. Consider following databases and draw ER diagram and convert entities

and relationships to relation table for a given scenario:

COMPANY DATABASE

EMPLOYEE (SSN, Name, Address, Salary, Sex, SuperSSN, DNO)
DEPARTMENT (DNo, DName, MgrSSN, MgrStartDate)
PROJECT (PNo, PName, PLocation, DNO)
DLOCATION (DNo, DLoc)
WORKS_ON (SSN, PNo, Hours)
MANAGES (SSN, DNo)

Solution:

1. The entities **EMPLOYEE, DEPARTMENT, PROJECT, and DLOCATION** are identified and represented using rectangles.
2. The attributes are represented using ellipses:

o **EMPLOYEE:** SSN (Primary Key), Name, Address, Salary, Sex, SuperSSN, DNO
♣ SSN is underlined as it is the Primary Key.
♣ SuperSSN is a foreign key referencing EMPLOYEE (self-relationship).
♣ DNO is a foreign key referencing DEPARTMENT.

o **DEPARTMENT:** DNo (Primary Key), DName, MgrSSN, MgrStartDate
♣ DNo is underlined as the Primary Key.
♣ MgrSSN is a foreign key referencing EMPLOYEE.

o **PROJECT:** PNo (Primary Key), PName, PLocation, DNO
♣ PNo is underlined as the Primary Key.
♣ DNO is a foreign key referencing DEPARTMENT.

o **DLOCATION:** DNo, DLoc
♣ DNo and DLoc together form a composite key.
♣ DNo is a foreign key referencing DEPARTMENT.

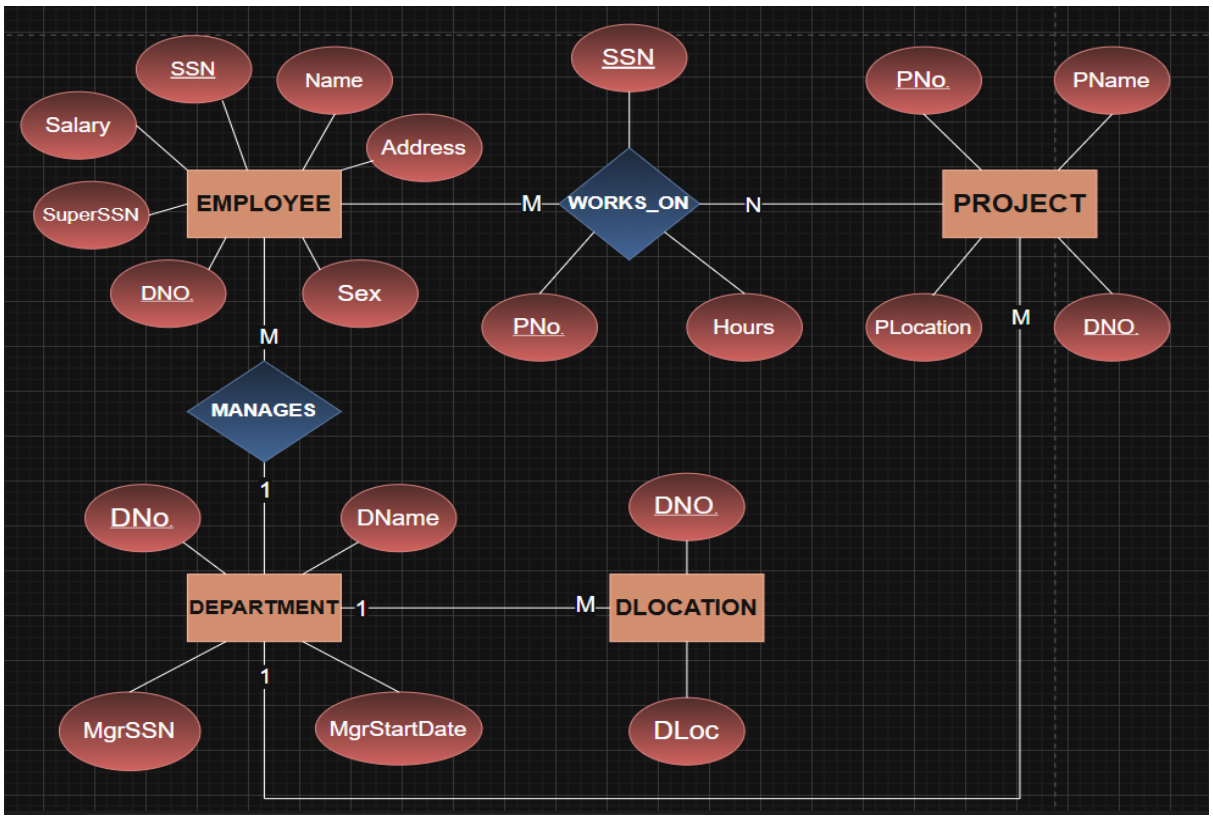
o **WORKS_ON:** SSN, PNo, Hours
♣ SSN and PNo together form a composite key.
♣ Hours is a descriptive attribute of the relationship.

3. Primary keys are represented by underlining the attribute names.
4. The relationship **WORKS_ON** is drawn as a diamond between **EMPLOYEE** and **PROJECT** with **M : N cardinality**, indicating that:
    - One employee can work on many projects.
    - One project can have many employees.

- Hours attribute represents the number of hours worked.

5. The relationship **MANAGES** is drawn between **EMPLOYEE** and **DEPARTMENT** with **1 : 1 cardinality**, indicating:
    - One employee manages one department.
    - Each department has one manager.
6. The relationship between **EMPLOYEE and DEPARTMENT** (via DNO) is **M : 1**, meaning:
    - Many employees work in one department.
7. The relationship between **DEPARTMENT and PROJECT** is **1 : M**, meaning:
    - One department controls many projects.
    - Each project belongs to one department.
8. The relationship between **DEPARTMENT and DLOCATION** is **1 : M**, meaning:
    - One department can have multiple locations.
9. A recursive relationship exists in EMPLOYEE through **SuperSSN**, representing that:
    - One employee can supervise many employees.
10. All entities, attributes, relationships, primary keys, foreign keys, and cardinalities are clearly represented in the ER diagram.

ER DIAGRAM:



Experiment : 3

Question : Database Schema for a customer-sale scenario

Customer(Cust id : integer, cust_name: string)

Item(item_id: integer, item_name: string, price: integer)

Sale(bill_no:integer,bill_data:date,cust_id: integer, item_id: integer, qty_sold: integer)

For the above schema, perform the following—

1. Create the tables with the appropriate integrity constraints.

Output:

```sql
CREATE DATABASE CUSTOMER_SALE;
USE CUSTOMER_SALE;
CREATE TABLE CUSTOMER(
CUST_ID INT PRIMARY KEY,
CUST_NAME VARCHAR(100)
);
CREATE TABLE ITEM (
ITEM_ID INT PRIMARY KEY,
ITEM_NAME VARCHAR(100),
PRICE INT);
CREATE TABLE SALE (
BILL_NO INT ,
BILL_DATA DATE,
CUST_ID INT,
ITEM_ID INT,
QTY_SOLD INT ,
FOREIGN KEY (CUST_ID) REFERENCES CUSTOMER(CUST_ID),
FOREIGN KEY (ITEM_ID) REFERENCES ITEM(ITEM_ID)
);
```

2. Insert around 10 records in each of the tables.

```sql
INSERT INTO
CUSTOMER(CUST_ID ,CUST_NAME)
VALUES
(1,'Krish'),
(2,'Harsh'),
(3,'Ankit'),
(4,'Aryan'),
(5,'Hridyanshu'),
(6,'Ashok'),
(7,'Alok'),
(8,'Aman'),
(9,'Ashish'),
(10,'Mohit');
```

CUST_ID	CUST_NAME
1	Krish
2	Harsh
3	Ankit
4	Aryan
5	Hridyanshu
6	Ashok
7	Alok
8	Aman
9	Ashish
10	Mohit
NULL	NULL

```sql
INSERT INTO ITEM VALUES
(101,'Pen',50),
(102,'Notebook',120),
(103,'Bag',800),
(104,'Bottle',250),
(105,'Shoes',1500),
(106,'Watch',2200),
(107,'Mouse',500),
(108,'Keyboard',700),
(109,'Charger',300),
(110,'Headphones',1800);
SELECT * FROM ITEM;
```

ITEM_ID	ITEM_NAME	PRICE
101	Pen	50
102	Notebook	120
103	Bag	800
104	Bottle	250
105	Shoes	1500
106	Watch	2200
107	Mouse	500
108	Keyboard	700
109	Charger	300
110	Headphones	1800
NULL	NULL	NULL

```sql
INSERT INTO SALE VALUES
(1,CURDATE(),1,103,1),
(2,CURDATE(),2,101,3),
(3,CURDATE(),2,105,2),
(4,'2026-02-15',3,104,1),
(5,'2026-02-14',4,106,1),
(6,'2026-02-13',5,102,4),
(7,CURDATE(),6,110,1),
(8,'2026-02-16',7,108,2),
(9,CURDATE(),5,109,5),
(10,'2026-02-12',8,107,1);
SELECT * FROM SALE;
```

BILL_NO	BILL_DATA	CUST_ID	ITEM_ID	QTY_SOLD
1	2026-02-23	1	103	1
2	2026-02-23	2	101	3
3	2026-02-23	2	105	2
4	2026-02-15	3	104	1
5	2026-02-14	4	106	1
6	2026-02-13	5	102	4
7	2026-02-23	6	110	1
8	2026-02-16	7	108	2
9	2026-02-23	5	109	5
10	2026-02-12	8	107	1

3.List all the bills for the current date with the customer names and item numbers

```sql
SELECT S.BILL_NO, C.CUST_NAME, S.ITEM_ID
FROM SALE S
JOIN CUSTOMER C ON S.CUST_ID = C.CUST_ID
WHERE S.BILL_DATA = CURDATE();
```

BILL_NO	CUST_NAME	ITEM_ID
1	Krish	103
2	Harsh	101
3	Harsh	105
7	Ashok	110
9	Hridyanshu	109

4. List the total Bill details with the quantity sold, price of the item and the final amount.

```
SELECT
    S.BILL_NO,
    C.CUST_NAME,
    I.ITEM_NAME,
    S.QTY_SOLD,
    I.PRICE,
    (S.QTY_SOLD * I.PRICE) AS FINAL_AMOUNT
FROM SALE S
JOIN CUSTOMER C ON S.CUST_ID = C.CUST_ID
JOIN ITEM I ON S.ITEM_ID = I.ITEM_ID;
```

BILL_NO	CUST_NAME	ITEM_NAME	QTY_SOLD	PRICE	FINAL_AMOUNT
1	Krish	Bag	1	800	800
2	Harsh	Pen	3	50	150
3	Harsh	Shoes	2	1500	3000
4	Ankit	Bottle	1	250	250
5	Aryan	Watch	1	2200	2200
6	Hridyanshu	Notebook	4	120	480
7	Ashok	Headphones	1	1800	1800
8	Alok	Keyboard	2	700	1400
9	Hridyanshu	Charger	5	300	1500
10	Aman	Mouse	1	500	500

5.List the details of the customer who have bought a product which has a price>200

```
SELECT DISTINCT C.*
FROM CUSTOMER C
JOIN SALE S ON C.CUST_ID = S.CUST_ID
JOIN ITEM I ON S.ITEM_ID = I.ITEM_ID
WHERE I.PRICE > 200;
```

CUST_ID	CUST_NAME
1	Krish
3	Ankit
2	Harsh
4	Aryan
8	Aman
7	Alok
5	Hridyanshu
6	Ashok

6.Give a count of how many products have been bought by each customer

```
SELECT C.CUST_NAME, SUM(S.QTY_SOLD) AS TOTAL_PRODUCTS_BOUGHT
FROM CUSTOMER C
JOIN SALE S ON C.CUST_ID = S.CUST_ID
GROUP BY C.CUST_NAME;
```

CUST_NAME	TOTAL_PRODUCTS_BOUGHT
Krish	1
Harsh	5
Ankit	1
Aryan	1
Hridyanshu	9
Ashok	1
Alok	2
Aman	1

7.Give a list of products bought by a customer having cust_id as 5.

```
SELECT I.ITEM_NAME, S.QTY_SOLD
FROM SALE S
JOIN ITEM I ON S.ITEM_ID = I.ITEM_ID
WHERE S.CUST_ID = 5;
```

ITEM_NAME	QTY_SOLD
Notebook	4
Charger	5

8.List the item details which are sold as of today.

```
SELECT I.*
FROM SALE S
JOIN ITEM I ON S.ITEM_ID = I.ITEM_ID
WHERE S.BILL_DATA = CURDATE();
```

ITEM_ID	ITEM_NAME	PRICE
103	Bag	800
101	Pen	50
105	Shoes	1500
110	Headphones	1800
109	Charger	300

9. Create a view which lists out the bill_no, bill_date, cust_id, item_id, price, qty_sold,Amount

```
CREATE VIEW Bill_Details AS
SELECT
    S.BILL_NO,
    S.BILL_DATA,
    S.CUST_ID,
    S.ITEM_ID,
    I.PRICE,
    S.QTY_SOLD,
    (S.QTY_SOLD * I.PRICE) AS AMOUNT
FROM SALE S
JOIN ITEM I ON S.ITEM_ID = I.ITEM_ID;

SELECT * FROM BILL_DETAILS;
```

BILL_NO	BILL_DATA	CUST_ID	ITEM_ID	PRICE	QTY_SOLD	AMOUNT
1	2026-02-23	1	103	800	1	800
2	2026-02-23	2	101	50	3	150
3	2026-02-23	2	105	1500	2	3000
4	2026-02-15	3	104	250	1	250
5	2026-02-14	4	106	2200	1	2200
6	2026-02-13	5	102	120	4	480
7	2026-02-23	6	110	1800	1	1800
8	2026-02-16	7	108	700	2	1400
9	2026-02-23	5	109	300	5	1500
10	2026-02-12	8	107	500	1	500

10.Create a view which lists the daily sales date wise for the last one week

```
CREATE VIEW WEEKLY_SALES AS
SELECT
    BILL_DATA,
    SUM(S.QTY_SOLD * I.PRICE) AS TOTAL_DAILY_SALES
FROM SALE S
JOIN ITEM I ON S.ITEM_ID = I.ITEM_ID
WHERE BILL_DATA >= CURDATE() - INTERVAL 7 DAY
GROUP BY BILL_DATA;

SELECT * FROM WEEKLY_SALES;
```

BILL_DATA	TOTAL_DAILY_SALES
2026-02-23	7250
2026-02-16	1400

Experiment : 4

Question : Consider the following table: Table: CLASS (ID,Name)

```
CREATE DATABASE KRISH;
USE KRISH;
CREATE TABLE CLASS(
ID INT,
NAME VARCHAR(100));
INSERT INTO
CLASS(ID ,NAME)
VALUES
(1,'Krish'),
(2,'Harsh'),
(3,'Ankit'),
(4,'Aryan'),
(5,'Hridyanshu'),
(6,'Ashok'),
(7,'Alok'),
(8,'Aman'),
(9,'Ashish'),
(10,'Mohit');
SELECT * FROM CLASS;
```

ID	NAME
1	Krish
2	Harsh
3	Ankit
4	Aryan
5	Hridyanshu
6	Ashok
7	Alok
8	Aman
9	Ashish
10	Mohit

1.SET AUTOCOMMIT = 0; and START TRANSACTION;

```
SET autocommit = 0;
START TRANSACTION;
```

2.INSERT INTO class VALUES (11,'Rahul');

```
INSERT INTO CLASS VALUES (11,'RAHUL');
SELECT * FROM CLASS;
```

ID	NAME
1	Krish
2	Harsh
3	Ankit
4	Aryan
5	Hridyanshu
6	Ashok
7	Alok
8	Aman
9	Ashish
10	Mohit
11	RAHUL
12	CHRIS
13	BRAVO
11	RAHUL

3.UPDATE class SET name = ‘Abhijeet’ WHERE id= ‘5’;

```
26 •  SET SQL_SAFE_UPDATES = 0;
27 •  UPDATE CLASS SET NAME = 'ABHIJEET' WHERE ID = 5;
28 •  SELECT * FROM CLASS;
```

ID	NAME
1	Krish
2	Harsh
3	Ankit
4	Aryan
5	ABHIJEET
6	Ashok
7	Alok
8	Aman
9	Ashish
10	Mohit
11	RAHUL

4. SAVEPOINT A:

```
28 •  SAVEPOINT A;
```

5. INSERT INTO class VALUES (12, ‘Chris’);

```
30 •  INSERT INTO CLASS VALUES (12,'CHRIS');
```

ID	NAME
1	Krish
2	Harsh
3	Ankit
4	Aryan
5	ABHIJEET
6	Ashok
7	Alok
8	Aman
9	Ashish
10	Mohit
11	RAHUL
12	CHRIS

6. SAVEPOINT B;

```
31 •  SAVEPOINT B;
```

7. INSERT INTO class VALUES (13, ‘Bravo’);

```
32 •  INSERT INTO CLASS VALUES (13,'BRAVO');
33 •  SELECT * FROM CLASS;
```

ID	NAME
1	Krish
2	Harsh
3	Ankit
4	Aryan
5	ABHIJEET
6	Ashok
7	Alok
8	Aman
9	Ashish
10	Mohit
11	RAHUL
12	CHRIS
13	BRAVO

8.SAVEPOINT C;

```
34 •    SAVEPOINT C;
```

9. ROLLBACK TO B;

```
36 •    ROLLBACK TO B;
```

	ID	NAME
▶	1	Krish
	2	Harsh
	3	Ankit
	4	Aryan
	5	ABHIJEET
	6	Ashok
	7	Alok
	8	Aman
	9	Ashish
	10	Mohit
	11	RAHUL
	12	CHRIS

10.ROLLBACK TO A;

```
38 •    ROLLBACK TO A;
39 •    SELECT * FROM CLASS;
```

	ID	NAME
▶	1	Krish
	2	Harsh
	3	Ankit
	4	Aryan
	5	ABHIJEET
	6	Ashok
	7	Alok
	8	Aman
	9	Ashish
	10	Mohit
	11	RAHUL

11. COMMIT;

```
40 •    COMMIT;
```

Experiment : 4.2

Consider the following table: Table: ACCOUNT_TXN

Acc_ID Holder_Name Balance

101	**Aman**	**5000**
102	**Bhavna**	**7000**
103	**Chirag**	**6000**

```
CREATE TABLE ACCOUNT_TXN(
ACC_ID INT,
HOLDER_NAME VARCHAR(100),
BALANCE INT
);
INSERT INTO ACCOUNT_TXN
VALUES
(101,'AMAN',5000),
(102,'BHAVNA',7000),
(103,'CHIRAG',6000);
SELECT * FROM ACCOUNT_TXN;
```

ACC_ID	HOLDER_NAME	BALANCE
101	AMAN	5000
102	BHAVNA	7000
103	CHIRAG	6000

Perform the following TCL-based SQL script and write the output after each SELECT statement:

1.INSERT INTO ACCOUNT_TXN VALUES (104, 'Diya', 4000);

```
INSERT INTO ACCOUNT_TXN
VALUES
(104,'DIYA',4000);
```

ACC_ID	HOLDER_NAME	BALANCE
101	AMAN	5000
102	BHAVNA	7000
103	CHIRAG	6000
104	DIYA	4000

2.SAVEPOINT S1

```
SET autocommit = 0;
START TRANSACTION;
SAVEPOINT S1;
```

3. UPDATE ACCOUNT_TXN SET Balance = Balance + 1500 WHERE Acc_ID = 101

```
UPDATE ACCOUNT_TXN SET BALANCE = BALANCE+1500 WHERE ACC_ID = 101;
SELECT * FROM ACCOUNT_TXN;
```

ACC_ID	HOLDER_NAME	BALANCE
101	AMAN	6500
102	BHAVNA	7000
103	CHIRAG	6000
104	DIYA	4000

4. UPDATE ACCOUNT_TXN SET Balance = Balance - 1000 WHERE Acc_ID = 102

```
UPDATE ACCOUNT_TXN SET BALANCE = BALANCE - 1000 WHERE ACC_ID = 102;
SELECT * FROM ACCOUNT_TXN;
```

ACC_ID	HOLDER_NAME	BALANCE
101	AMAN	8000
102	BHAVNA	6000
103	CHIRAG	6000
104	DIYA	4000

5. SAVEPOINT S2;

```
SAVEPOINT S2;
```

6. DELETE FROM ACCOUNT_TXN WHERE Acc_ID = 103

```
DELETE FROM ACCOUNT_TXN WHERE ACC_ID = 103;
SELECT * FROM ACCOUNT_TXN;
```

ACC_ID	HOLDER_NAME	BALANCE
101	AMAN	8000
102	BHAVNA	6000
104	DIYA	4000

7. INSERT INTO ACCOUNT_TXN VALUES (105, 'Eshan', 3000);

```
INSERT INTO ACCOUNT_TXN
VALUES
(105,'ESHAN',3000);
SELECT * FROM ACCOUNT_TXN;
```

ACC_ID	HOLDER_NAME	BALANCE
101	AMAN	8000
102	BHAVNA	6000
104	DIYA	4000
105	ESHAN	3000

8.SAVEPOINT S3

```
SAVEPOINT S3;
```

9. UPDATE ACCOUNT_TXN SET Holder_Name = 'Bhavna Sharma' WHERE Acc_ID = 102;

```
UPDATE ACCOUNT_TXN SET HOLDER_NAME = 'BHAVNA SHARMA' WHERE ACC_ID = 102;
SELECT * FROM ACCOUNT_TXN;
```

ACC_ID	HOLDER_NAME	BALANCE
101	AMAN	5000
102	BHAVNA SHARMA	7000
103	CHIRAG	6000
104	DIYA	4000

11.ROLLBACK TO S2;

```
ROLLBACK TO S2;
```

ACC_ID	HOLDER_NAME	BALANCE
101	AMAN	6500
102	BHAVNA SHARMA	6000
103	CHIRAG	6000
104	DIYA	4000

12.INSERT INTO ACCOUNT_TXN VALUES (106, 'Farah', 4500)

```
INSERT INTO ACCOUNT_TXN
VALUES
(106,'FARAH',4500);
SELECT * FROM ACCOUNT_TXN;
```

ACC_ID	HOLDER_NAME	BALANCE
101	AMAN	6500
102	BHAVNA SHARMA	6000
103	CHIRAG	6000
104	DIYA	4000
104	DIYA	4000
106	FARAH	4500

13.SAVEPOINT S4;

```
SAVEPOINT S4;
```

14.UPDATE ACCOUNT_TXN SET Balance = Balance + 500 WHERE Acc_ID IN (101,104);

SELECT * FROM ACCOUNT_TXN

```
UPDATE ACCOUNT_TXN SET BALANCE = BALANCE + 500 WHERE ACC_ID IN (101,104);
SELECT * FROM ACCOUNT_TXN;
```

ACC_ID	HOLDER_NAME	BALANCE
101	AMAN	7000
102	BHAVNA SHARMA	6000
103	CHIRAG	6000
104	DIYA	4500
104	DIYA	4500
106	FARAH	4500

15.ROLLBACK TO S4;

SELECT * FROM ACCOUNT_TXN;

```
ROLLBACK TO S4;
SELECT * FROM ACCOUNT_TXN;
```

ACC_ID	HOLDER_NAME	BALANCE
101	AMAN	6500
102	BHAVNA SHARMA	6000
103	CHIRAG	6000
104	DIYA	4000
104	DIYA	4000
106	FARAH	4500

4000

16.COMMIT

```
COMMIT;
```

17.Identify which rows are permanently saved after COMMIT.

	ACC_ID	HOLDER_NAME	BALANCE
▶	101	AMAN	6500
	102	BHAVNA SHARMA	6000
	103	CHIRAG	6000
	104	DIYA	4000
	104	DIYA	4000
	106	FARAH	4500

Assignment 5

EMPLOYEES (Employee_Id, First_Name, Last_Name, Email, Phone_Number, Hire_Date, Job_Id, Salary, Commission_Act, Manager_Id, Department_Id)

(a) Find out the employee id, names, salaries of all the employees

(b) List out the employees who works under manager 100

(c) Find the names of the employees who have a salary greater than or equal to 4800

(d) List out the employees whose last name is 'AUSTIN'

(e) Find the names of the employees who works in departments 60,70 and 80 (f) Display the unique Manager_Id.

Query :

```sql
CREATE TABLE EMPLOYEES(
    Employee_Id INT PRIMARY KEY,
    First_Name VARCHAR(30),
    Last_Name VARCHAR(30),
    Email VARCHAR(50),
    Phone_Number VARCHAR(20),
    Hire_Date DATE,
    Job_Id VARCHAR(20),
    Salary INT,
    Commission_Pct FLOAT,
    Manager_Id INT,
    Department_Id INT
);
```

Data :

```
INSERT INTO EMPLOYEES VALUES
(101,'KRISH','SINGHAL','krish@gmail.com','9876543210','2023-01-10','SOFTWARE_DEV',12000,NULL,100,60);
INSERT INTO EMPLOYEES VALUES
(102,'HARSH','VARDHAN','harsg@gmail.com','9876543211','2023-02-12','HR_REP',4800,NULL,101,70);
INSERT INTO EMPLOYEES VALUES
(103,'HRIDYANSHU','SINGH','hridyanshu@gmail.com','9876543212','2023-03-15','SA_REP',5200,0.10,100,80);
INSERT INTO EMPLOYEES VALUES
(104,'ANKIT','SINGH','abkit@gmail.com','9876543213','2023-04-18','SOFTWARE_DEV',4500,NULL,102,60);
INSERT INTO EMPLOYEES VALUES
(105,'NIRVIKALP','CHAIDHARy','nirvikalp@gmail.com','9876543214','2023-05-20','SA_REP',7000,0.15,100,80);
INSERT INTO EMPLOYEES VALUES
(106,'ROHAN','YADAV','rohan@gmail.com','9876543215','2023-06-22','HR_REP',5000,NULL,103,70);
```

1.) **Find out the employee id, names, and salaries of all the employees..**

```
SELECT Employee_Id, First_Name, Last_Name, Salary
FROM EMPLOYEES;
```

Output :

Employee_Id	First_Name	Last_Name	Salary
101	KRISH	SINGHAL	12000
102	HARSH	VARDHAN	4800
103	HRIDYANSHU	SINGH	5200
104	ANKIT	SINGH	4500
105	NIRVIKALP	CHAIDHARy	7000
106	ROHAN	YADAV	5000
NULL	NULL	NULL	NULL

2.)List out the employees who work under manager 100.

```
SELECT *
FROM EMPLOYEES
WHERE Manager_Id = 100;
```

Output:

Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Commission_Pct	Manager_Id	Department_Id
101	KRISH	SINGHAL	krish@gmail.com	9876543210	2023-01-10	SOFTWARE_DEV	12000	NULL	100	60
103	HRIDYANSHU	SINGH	hridyanshu@gmail.com	9876543212	2023-03-15	SA_REP	5200	0.1	100	80
105	NIRVIKALP	CHAIDHARy	nirvikalp@gmail.com	9876543214	2023-05-20	SA_REP	7000	0.15	100	80
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

3.)Find the names of the employees who have a salary greater than or equal to 4800.

```
SELECT First_Name, Last_Name
FROM EMPLOYEES
WHERE Salary >= 4800;
```

Output:

	First_Name	Last_Name
▶	KRISH	SINGHAL
	HARSH	VARDHAN
	HRIDYANSHU	SINGH
	NIRVIKALP	CHAIDHARy
	ROHAN	YADAV

4.)List out the employees whose last name is 'AUSTIN'.

```
SELECT *
FROM EMPLOYEES
WHERE Last_Name = 'Austin';
```

Output:

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Commission_Pct	Manager_Id	Department_Id
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

5.)Find the names of the employees who work in departments 60, 70, and 80.

```
SELECT First_Name, Last_Name
FROM EMPLOYEES
WHERE Department_Id IN (60,70,80);
```

Output:

	First_Name	Last_Name
▶	KRISH	SINGHAL
	HARSH	VARDHAN
	HRIDYANSHU	SINGH
	ANKIT	SINGH
	NIRVIKALP	CHAIDHARy
	ROHAN	YADAV

6.)Display the unique Manager_Id.

```
SELECT DISTINCT Manager_Id
FROM EMPLOYEES;
```

Output:

Result Grid

	Manager_Id
▶	100
	101
	102
	103

Experiment : 6

Question : Create Client_master with the following fields (ClientNO, Name, Address, City, State, bal_due)

(a) Insert five records

(b) Find the names of clients whose bal_due> 5000 .

(c) Change the bal_due of ClientNO " C123" to Rs. 5100

(d) Change the name of Client_master to Client12.

(e) Display the bal_due heading as "BALANCE"

Query :

```
CREATE TABLE CLIENT_MASTER (
    CLIENTNO VARCHAR(10),
    NAME VARCHAR(50),
    ADDRESS VARCHAR(100),
    CITY VARCHAR(50),
    STATE VARCHAR(50),
    BAL_DUE DECIMAL(10,2)
);
```

a) Insert five records

```
INSERT INTO CLIENT_MASTER VALUES ('C101', 'Krish Singhal', 'MG ROAD', 'DELHI', 'DELHI', 6000);
INSERT INTO CLIENT_MASTER VALUES ('C102', 'Harsh Bardhan', 'BRIGADE ROAD', 'BANGALORE', 'KARNATAKA', 4500);
INSERT INTO CLIENT_MASTER VALUES ('C123', 'Ankit', 'PARK STREET', 'KOLKATA', 'WEST BENGAL', 3000);
INSERT INTO CLIENT_MASTER VALUES ('C104', 'Hridyanshu', 'FC ROAD', 'PUNE', 'MAHARASHTRA', 7000);
INSERT INTO CLIENT_MASTER VALUES ('C105', 'Rohan', 'ASHOK NAGAR', 'CHENNAI', 'TAMIL NADU', 2000);
```

Output:

CLIENTNO	NAME	ADDRESS	CITY	STATE	BAL_DUE
C101	Krish Singhal	MG ROAD	DELHI	DELHI	6000.00
C102	Harsh Bardhan	BRIGADE ROAD	BANGALORE	KARNATAKA	4500.00
C123	Ankit	PARK STREET	KOLKATA	WEST BENGAL	3000.00
C104	Hridyanshu	FC ROAD	PUNE	MAHARASHTRA	7000.00
C105	Rohan	ASHOK NAGAR	CHENNAI	TAMIL NADU	2000.00

(b) Find the names of clients whose bal_due> 5000 .

```
SELECT NAME
FROM CLIENT_MASTER
WHERE BAL_DUE > 5000;
```

NAME
Krish Singhal
Hridyanshu

(c) Change the bal_due of ClientNO " C123" to Rs. 5100

```
SET SQL_SAFE_UPDATES = 0;
UPDATE CLIENT_MASTER
SET BAL_DUE = 5100
WHERE CLIENTNO = 'C123';
SELECT * FROM CLIENT_MASTER;
```

Output :

CLIENTNO	NAME	ADDRESS	CITY	STATE	BAL_DUE
C101	Krish Singhal	MG ROAD	DELHI	DELHI	6000.00
C102	Harsh Bardhan	BRIGADE ROAD	BANGALORE	KARNATAKA	4500.00
C123	Ankit	PARK STREET	KOLKATA	WEST BENGAL	5100.00
C104	Hridyanshu	FC ROAD	PUNE	MAHARASHTRA	7000.00
C105	Rohan	ASHOK NAGAR	CHENNAI	TAMIL NADU	2000.00

(d) Change the name of Client_master to Client12.

```
ALTER TABLE CLIENT_MASTER
RENAME TO CLIENT12;
SELECT * FROM CLIENT_MASTER;
SELECT * FROM CLIENT12;
```

Output:

CLIENTNO	NAME	ADDRESS	CITY	STATE	BAL_DUE
C101	Krish Singhal	MG ROAD	DELHI	DELHI	6000.00
C102	Harsh Bardhan	BRIGADE ROAD	BANGALORE	KARNATAKA	4500.00
C123	Ankit	PARK STREET	KOLKATA	WEST BENGAL	5100.00
C104	Hridyanshu	FC ROAD	PUNE	MAHARASHTRA	7000.00
C105	Rohan	ASHOK NAGAR	CHENNAI	TAMIL NADU	2000.00

(e) Display the bal_due heading as “BALANCE”

```
SELECT BAL_DUE AS BALANCE
FROM CLIENT12;
```

Output:

BALANCE
6000.00
4500.00
5100.00
7000.00
2000.00

ASSIGNMENT : 7

Rollback and Commit commands Create Teacher table with the following fields (Name, DeptNo, Date of joining, DeptName, Location, Salary)

```
CREATE DATABASE ASSIGNEMENT6;
USE ASSIGNEMENT6;
CREATE TABLE Teacher (
    Name VARCHAR(50),
    DeptNo INT,
    DateOfJoining DATE,
    DeptName VARCHAR(50),
    Location VARCHAR(50),
    Salary DECIMAL(10,2)
);
```

(a) Insert five records

```
INSERT INTO Teacher VALUES
('Krish', 1, '2020-06-15', 'Mathematics', 'Delhi', 50000),
 ('Harsh', 2, '2019-03-10', 'Commerce', 'Mumbai', 45000),
 ('Raj', 1, '2018-07-20', 'Mathematics', 'Delhi', 55000),
 ('Priya', 3, '2021-01-05', 'Science', 'Chennai', 40000),
('Karan', 2, '2017-09-25', 'Commerce', 'Mumbai', 48000);
```

```
SELECT * FROM Teacher;
```

Name	DeptNo	DateOfJoining	DeptName	Location	Salary
Krish	1	2020-06-15	Mathematics	Delhi	50000.00
Harsh	2	2019-03-10	Commerce	Mumbai	45000.00
Raj	1	2018-07-20	Mathematics	Delhi	55000.00
Priya	3	2021-01-05	Science	Chennai	40000.00
Karan	2	2017-09-25	Commerce	Mumbai	48000.00

(b) Give Increment of 25% salary for Mathematics Department .

```
START TRANSACTION;
SET SQL_SAFE_UPDATES = 0;
UPDATE Teacher
SET Salary = Salary + (Salary * 0.25)
WHERE DeptName = 'Mathematics';
SELECT * FROM Teacher;
```

	Name	DeptNo	DateOfJoining	DeptName	Location	Salary
▶	Krish	1	2020-06-15	Mathematics	Delhi	62500.00
	Harsh	2	2019-03-10	Commerce	Mumbai	45000.00
	Raj	1	2018-07-20	Mathematics	Delhi	68750.00
	Priya	3	2021-01-05	Science	Chennai	40000.00
	Karan	2	2017-09-25	Commerce	Mumbai	48000.00

(c) Perform Rollback command

```
ROLLBACK;
SELECT * FROM Teacher;
```

	Name	DeptNo	DateOfJoining	DeptName	Location	Salary
▶	Krish	1	2020-06-15	Mathematics	Delhi	50000.00
	Harsh	2	2019-03-10	Commerce	Mumbai	45000.00
	Raj	1	2018-07-20	Mathematics	Delhi	55000.00
	Priya	3	2021-01-05	Science	Chennai	40000.00
	Karan	2	2017-09-25	Commerce	Mumbai	48000.00

(d)GiveIncrementof15%salaryforCommerce Department

```
UPDATE Teacher
SET Salary = Salary + (Salary * 0.15)
WHERE DeptName = 'Commerce';
SELECT * FROM Teacher;
```

	Name	DeptNo	DateOfJoining	DeptName	Location	Salary
▶	Krish	1	2020-06-15	Mathematics	Delhi	50000.00
	Harsh	2	2019-03-10	Commerce	Mumbai	51750.00
	Raj	1	2018-07-20	Mathematics	Delhi	55000.00
	Priya	3	2021-01-05	Science	Chennai	40000.00
	Karan	2	2017-09-25	Commerce	Mumbai	55200.00

(e) Perform commit command

```
COMMIT;
```

ASSIGNMENT : 8

(Exercise on order by and group by clauses) Create Sales table with the following fields (Sales No, Salesname, Branch, Salesamount, DOB)

```
CREATE TABLE Sales (
    SalesNo INT,
    SalesName VARCHAR(50),
    Branch VARCHAR(50),
    SalesAmount DECIMAL(10,2),
    DOB DATE
);
```

(a) Insert five records

```
INSERT INTO Sales VALUES
 (1, 'Krish', 'Delhi', 10000, '1995-12-21'),
 (2, 'Harsh', 'Mumbai', 15000, '1998-05-10'),
 (3, 'Raj', 'Delhi', 20000, '1992-12-11'),
 (4, 'Priya', 'Chennai', 12000, '1997-07-25'),
(5, 'Karan', 'Mumbai', 18000, '1994-12-05');
SELECT * FROM Sales;
```

	SalesNo	SalesName	Branch	SalesAmount	DOB
▶	1	Krish	Delhi	10000.00	1995-12-21
	2	Harsh	Mumbai	15000.00	1998-05-10
	3	Raj	Delhi	20000.00	1992-12-11
	4	Priya	Chennai	12000.00	1997-07-25
	5	Karan	Mumbai	18000.00	1994-12-05

(b) Calculate total salesamount in each branch

```
SELECT Branch, SUM(SalesAmount) AS TotalSales
FROM Sales
GROUP BY Branch;
```

	Branch	TotalSales
▶	Delhi	30000.00
	Mumbai	33000.00
	Chennai	12000.00

(c) Calculate average salesamount in each branch

```sql
SELECT Branch, AVG(SalesAmount) AS AvgSales
FROM Sales
GROUP BY Branch;
```

Branch	AvgSales
Delhi	15000.000000
Mumbai	16500.000000
Chennai	12000.000000

(d) Display all the salesmen, DOB who are born in the month of December as day in character format i.e. 21-Dec-09

```sql
SELECT SalesName, DATE_FORMAT(DOB, '%d-%b-%y') AS DOB
FROM Sales
WHERE MONTH(DOB) = 12;
```

SalesName	DOB
Krish	21-Dec-95
Raj	11-Dec-92
Karan	05-Dec-94

(e) Display the name and DOB of salesman in alphabetical order of the month.

```sql
SELECT SalesName, DOB
FROM Sales
ORDER BY DATE_FORMAT(DOB, '%M');
```

SalesName	DOB
Krish	1995-12-21
Raj	1992-12-11
Karan	1994-12-05
Priya	1997-07-25
Harsh	1998-05-10

ASSIGNMENT : 9

Consider the following tables namely “DEPARTMENTS” and “EMPLOYEES” Their schemas

are as follows,

Departments (dept no , dept name , dept_location); Employees (emp_id , emp_name , emp_salary,dept_no);

```
CREATE TABLE Departments (
    dept_no INT,
    dept_name VARCHAR(50),
    dept_location VARCHAR(50)
);

CREATE TABLE Employees (
    emp_id INT,
    emp_name VARCHAR(50),
    emp_salary DECIMAL(10,2),
    dept_no INT
);
```

a)Develop a query to grant all privileges of employees table into departments table

```
GRANT ALL PRIVILEGES ON Employees TO 'root'@'localhost';
```

b)Develop a query to grant some privileges of employees table into departments table

```
GRANT SELECT, INSERT ON Employees TO 'root'@'localhost';
```

c)Develop a query to revoke all privileges of employees table from departments table

```
CREATE USER 'student'@'localhost' IDENTIFIED BY '1234';
GRANT ALL PRIVILEGES ON Employees TO 'student'@'localhost';
REVOKE ALL PRIVILEGES ON Employees FROM 'student'@'localhost';
```

d)Develop a query to revoke some privileges of employees table from departments table

```
REVOKE SELECT, INSERT ON Employees FROM 'root'@'localhost';
```

e)Write a query to implement the save point.

```
START TRANSACTION;
SAVEPOINT sp1;
UPDATE Employees
SET emp_salary = emp_salary + 1000;
SELECT * FROM Employees;
```

	emp_id	emp_name	emp_salary	dept_no
▶	101	Krish	51000.00	1
	102	Harsh	61000.00	2
	103	Raj	56000.00	3
	104	Priya	46000.00	2
	105	Karan	71000.00	3
	106	Sneha	41000.00	4
	107	Vikas	66000.00	5

```
ROLLBACK TO sp1;
SELECT * FROM Employees;
```

	emp_id	emp_name	emp_salary	dept_no

```
COMMIT;
```

ASSIGNMENT : 10

Using the tables “DEPARTMENTS” and “EMPLOYEES” perform the following queries

```sql
CREATE DATABASE ASS10;
USE ASS10;
CREATE TABLE Departments (
    dept_no INT,
    dept_name VARCHAR(50),
    dept_location VARCHAR(50)
);
INSERT INTO Departments VALUES
(1, 'HR', 'Delhi'),
(2, 'Finance', 'Mumbai'),
(3, 'IT', 'Bangalore'),
(4, 'Marketing', 'Chennai'),
(5, 'Sales', 'Hyderabad');
```

```sql
CREATE TABLE Employees (
    emp_id INT,
    emp_name VARCHAR(50),
    emp_salary DECIMAL(10,2),
    dept_no INT
);
INSERT INTO Employees VALUES
(101, 'Krish', 50000, 1),
(102, 'Harsh', 60000, 2),
(103, 'Raj', 55000, 3),
(104, 'Priya', 45000, 2),
(105, 'Karan', 70000, 3),
(106, 'Sneha', 40000, 4),
(107, 'Vikas', 65000, 5);
```

a)Display the employee details, departments that the departments are same in both the emp and dept.

```sql
SELECT *
FROM Employees e
INNER JOIN Departments d
ON e.dept_no = d.dept_no;
```

	emp_id	emp_name	emp_salary	dept_no	dept_no	dept_name	dept_location
▶	101	Krish	50000.00	1	1	HR	Delhi
	102	Harsh	60000.00	2	2	Finance	Mumbai
	103	Raj	55000.00	3	3	IT	Bangalore
	104	Priya	45000.00	2	2	Finance	Mumbai
	105	Karan	70000.00	3	3	IT	Bangalore
	106	Sneha	40000.00	4	4	Marketing	Chennai
	107	Vikas	65000.00	5	5	Sales	Hyderabad

b)Display the employee name and Department name by implementing a left outer join.

```
SELECT e.emp_name, d.dept_name
FROM Employees e
LEFT JOIN Departments d
ON e.dept_no = d.dept_no;
```

	emp_name	dept_name
▶	Krish	HR
	Harsh	Finance
	Raj	IT
	Priya	Finance
	Karan	IT
	Sneha	Marketing
	Vikas	Sales

c)Display the employee name and Department name by implementing a right outer join.

```
SELECT e.emp_name, d.dept_name
FROM Employees e
RIGHT JOIN Departments d
ON e.dept_no = d.dept_no;
```

	emp_name	dept_name
▶	Krish	HR
	Priya	Finance
	Harsh	Finance
	Karan	IT
	Raj	IT
	Sneha	Marketing
	Vikas	Sales

d)Display the details of those who draw the salary greater than the average salary

```
SELECT *
FROM Employees
WHERE emp_salary > (SELECT AVG(emp_salary) FROM Employees);
```

emp_id	emp_name	emp_salary	dept_no
102	Harsh	60000.00	2
105	Karan	70000.00	3
107	Vikas	65000.00	5